HexOct: Knitting Crystals

A knitted HexOct fabric: its construction, its mechanics and its comfort, for every close-packed stacking



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Paper #83 · Version 5 · 10 November 2026 (Version 4: 3 November 2026; Version 3: withdrawn for revision, 2 November 2026; Versions 1–2: 1 November 2026) · DICTO DOI: 10.5281/zenodo.23848271 (concept) · Licence CC BY-SA 4.0
The Knitted HexOct, its idea and its name are DICTO's. Design, computation, checks and drafting with Claude (Anthropic).

Version 5 corrects the notes on methods: the names of our computers and private file paths are removed. No result changes.

Abstract. DICTO proposed a fabric that is a crystal: one continuous multi-layer knitted fabric whose knitted sheets are the close-packed layers of balls and whose linking yarns join each node to the three nodes of the next sheet that its ball touches, turning on in the same sense at a c layer and back at an h layer. The cells it closes off are the pieces of the HexOct tiles (#75), so any close-packed stacking can be knitted into the fabric's own thickness. Part 1 gives the construction: geometry, yarn and stitch plan by hand and by machine, and two test pieces. Part 2 tests whether the stacking is a design dial at equal density. Part 3 (this front section) adds comfort: vapour transfer, conduction along the yarn and direct radiation.

Knit the layers of a crystal as layers of fabric and tie them together as the crystal is tied. Every way of stacking the layers weighs the same and squashes the same; what changes is how much you can see straight through it and how easily it shears.

Main result

Result. In the idealised Knitted HexOct, all close-packed stackings have the same density; through-thickness compression, buckling onset, rigidity, vapour transfer, conduction along the yarn and direct radiation through twelve sheets are the same in every stacking. The stacking changes only two things: the straight see-through area (up to 10.7 times larger in 2H than in 3C at link radius 0.05 a) and, when the sheets are much softer than the links, the free transverse shear (3C up to 17.5 times softer than 2H), which is set by the stack's net turning (its mean Hägg sum), not by the share of c layers.

Version history

Versions 1 and 2 knitted an edge-on slice of the stackings: a picture of the crystal on a flat fabric. DICTO proposed instead a fabric that is the crystal in three dimensions, and Version 3 withdrew the earlier version for that revision. Version 4 is the revision: the Knitted HexOct itself, specified and analysed. The fabric has not yet been knitted; the two test pieces of Part 1 (a c stack and an h stack, 27 nodes and 42 links each, about 210 m of DK cotton each) are the next step, and the knitted pieces will be added in a later version.

Contents

Front section: abstract, main result, version history, Part 3 (comfort). Part 1: *Knitted HexOct, construction specification* (5 pages). Part 2: *Knitted HexOct: is the c/h stacking a design dial?* (lattice mechanics and a see-through proxy, 4 pages).

Part 3. Comfort: vapour, conduction and radiation

On the same voxel model as Part 2 (members as capsules of radius r on the periodic cell, twelve sheets), we computed for seven stackings: the effective diffusivity of vapour through the air spaces (a Laplace solve, D_{eff}/D_{air}), conduction along the yarn alone (the same solve in the solid, k_{eff}/k_{yarn}), and the share of diffuse (cosine-weighted) rays that pass through all twelve sheets without touching a member.

Stacking (r = 0.05 a)	Porosity	D_{eff}/D_{air}	Tortuosity factor	k_{eff}/k_{yarn}	Direct radiation
3C	0.9395	0.8821	1.065	0.01852	0.14 %
2H	0.9395	0.8823	1.065	0.01852	0.14 %
4H	0.9395	0.8822	1.065	0.01852	0.04 %
6H	0.9395	0.8822	1.065	0.01852	0.03 %
Fibonacci	0.9395	0.8822	1.065	0.01852	0.03 %
Graded (c top, h bottom)	0.9395	0.8822	1.065	0.01852	0.03 %
Graded (h top, c bottom)	0.9395	0.8822	1.065	0.01852	0.02 %

> Table 1. Comfort proxies, grid 0.025 a (3C and 2H also at 0.0167 a: 0.8855 and 0.8856, conduction 0.0201). At r = 0.1 a the same holds: D_{eff}/D_{air} 0.640–0.641 and conduction 0.0898 in every stacking, direct radiation at most 0.04 %.

So in the idealised fabric the stacking does not change how well it breathes or insulates: vapour passes about equally well through every stacking (88 % of open air at this yarn thickness), conduction along the yarn is the same, and with twelve sheets almost no radiation passes straight through in any stacking, so the extra see-through area of 2H along the stacking axis does not carry over to diffuse radiation. Comfort is set by the yarn and the thickness, not by the stacking. The limits are those of Part 2: idealised straight members, no yarn mechanics, no moisture in the fibre, no air flow solved.

> **Data and code.** Studies 2026-11-02-knitted-hexoct (Part 1), 2026-11-02-hexoct-knit-mechanics (Part 2; scripts/therm.py and run_therm.sh for Part 3, results data/therm-summary.md), research branch of the kaleidohedra repository.

> **References.** DICTO, "The Rule of Halves: the Holes of Every Close-Packed Stacking", paper #57, DICTO DOI: 10.5281/zenodo.23647898 (2026). DICTO, "HexOct: Two Tiles for Every Stacking", paper #75, DICTO DOI: 10.5281/zenodo.23737639 (2026). DICTO, "The Window Law", paper #84, DICTO DOI: 10.5281/zenodo.23848832 (2026). Further references in Parts 1 and 2.

Knitted HexOct

Construction specification: one continuous multi-layer knitted fabric whose sheets and links are a close-packed stacking

DICTO · dictospheres.com · 2 November 2026 · Part 1 of *HexOct: Knitting Crystals*, version 4
· DICTO DOI: 10.5281/zenodo.23848271 (concept) · study 2026-11-02-knitted-hexoct

DICTO proposed the Knitted HexOct: one continuous multi-layer knitted fabric, built like a spacer fabric. Knitted sheets lie one above another; each sheet is one close-packed layer of balls, its nodes on a triangular lattice. Linking yarns join neighbouring sheets: each node is linked to the 3 nodes of the next sheet that its ball touches. Whether a layer is c or h is carried by the links: at a c sheet the links go on in the same turning sense as below (A→B→C), at an h sheet they turn back (A→B→A). The cells closed off by sheets and links are the pieces of the HexOct c-tiles and h-tiles (#75). Any stacking can be knitted, and the structure is in the fabric's own build through its thickness, not drawn on its surface.

Summary. All links have the length of the node spacing a, rise at 54.74° to the sheets, and hold the sheets $\sqrt{(2/3)}\ a$ apart. At a c sheet each link up continues a link down in one straight line; at an h sheet each link up is the mirror image of a link down in the sheet. Pulled apart, three taut links per node centre each sheet exactly over the one below, so the knitter only has to get the link *lengths* right and link the right nodes. The test piece is three sheets of 3 × 3 nodes, knitted twice: A-B-C (middle sheet c) and A-B-A (middle sheet h), 27 nodes and 42 links each. Best by hand: garter-stitch sheets with marked nodes, links as 3-stitch i-cords. By machine, one gap at a time is standard (double-needle-bar spacer); three or more spaced sheets in one run were not found in the sources checked.

1. Exact geometry

Our check: scripts/check.py, exact (sympy) Sheet k lies at height $k \cdot d$ with $\boldsymbol{d = \sqrt{(2/3)}\ a \approx 0.8165\ a}$. In every sheet the nodes form a triangular lattice of spacing a: node (i, j) sits at $i \cdot \mathbf{e}_1 + j \cdot \mathbf{e}_2$ from the sheet's origin, with $\mathbf{e}_1 = (a, 0)$ and $\mathbf{e}_2 = (a/2, \sqrt{3}\ a/2)$, so rows of nodes are $(\sqrt{3}/2)\ a \approx 0.866\ a$ apart. A sheet on position A has its origin at (0, 0), on B at $(\mathbf{e}_1+\mathbf{e}_2)/3 = (a/2, a/(2\sqrt{3}))$, on C at $2(\mathbf{e}_1+\mathbf{e}_2)/3$. Neighbouring sheets never share a position.

The three links from an A node to the B sheet have horizontal offsets

link	horizontal offset (plan)	direction in plan	target, in the upper sheet's own indices
1	$(+a/2, +a/(2\sqrt{3}))$	30°	(i, j)
2	$(-a/2, +a/(2\sqrt{3}))$	150°	$(i-1, j)$
3	$(0, -a/\sqrt{3})$	270°	$(i, j-1)$

Each offset has length $a/\sqrt{3}$, so each link has length $\sqrt{(a^2/3 + 2a^2/3)} = \boldsymbol{a}$**, equal to the node spacing**, and rises at arctan $\sqrt{2}$ = **54.7356°** above the sheet (35.26° from the vertical). Every step to the next letter (A→B, B→C, C→A, a *forward* step) uses these three offsets. Every step to the previous letter (B→A, C→B, A→C, a *backward* step) uses the three opposite offsets, at 210°, 330° and 90°, with targets (i, j), $(i+1, j)$, $(i, j+1)$. All twelve cases were checked exactly: length a, tilt 54.7356°.

c and h at the next level. A sheet is c when the steps below and above it are of the same kind (both forward, as B in A-B-C, or both backward) and h when they differ (B in A-B-A). At a node of a **c** sheet, each of its 3 links up points exactly opposite one of its 3 links down: the two form one straight line through the node (3 of 3 checked), so the bonds run straight on through the sheet. At a node of an **h** sheet, each link up is the mirror image, in the sheet plane, of one link down: the same plan direction, rising instead of falling (3 of 3 checked), so the bonds fold back. This is the exact meaning of "same turning sense" and "turn back". In the section seen along a row of nodes (Figure 1) a run of c sheets shows the links as straight slants and an h sheet as a chevron, the same slant-and-chevron law as the edge-on knitted slice of #83, here built into the thickness instead of charted on a face.

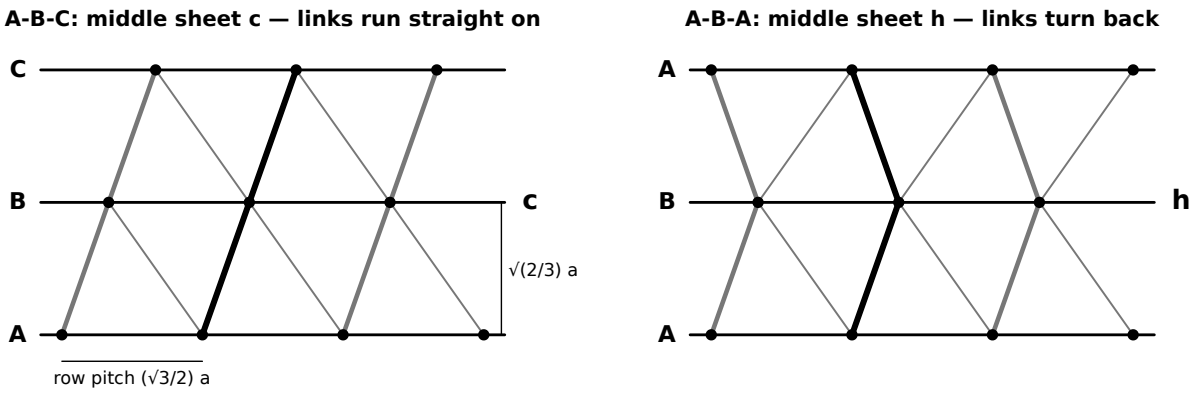


Figure 1. Construction section, seen along a row of nodes. Horizontal lines: sheets; dots: rows of nodes; slanted lines: links. A thick line is a pair of links, one in front of and one behind the drawing plane (true length *a*, shown shortened); a thin line is one link lying in the drawing plane (true length *a*). Rows of A, B, C sit at 0, 1/3 and 2/3 of the row pitch. The black line traces one pair of links through the middle sheet: straight on at a c sheet (left), turned back at an h sheet (right).

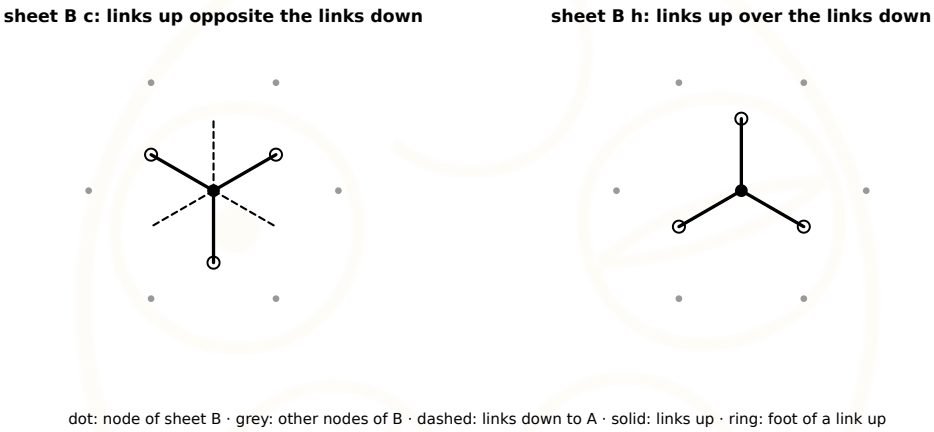


Figure 2. Plan at one node of the middle sheet B. Dashed: the 3 links down to sheet A (plan directions 330°, 90°, 210°). Solid: the 3 links up. c (A-B-C): links up at 150°, 270°, 30°, each opposite a link down. h (A-B-A): links up at 330°, 90°, 210°, each over a link down.

The links centre the sheets. Our check: one line of algebra, plus 20 000 random shifts If sheet B is moved sideways by $\mathbf{t}$ from its place, the three plan offsets $\mathbf{u}_1$, $\mathbf{u}_2$, $\mathbf{u}_3$ (length $a/\sqrt{3}$, sum zero) become $\mathbf{u}_m + \mathbf{t}$, and $|\mathbf{u}_m + \mathbf{t}|^2 = a^2/3 + 2\mathbf{u}_m\cdot\mathbf{t} + |\mathbf{t}|^2$. Since the $\mathbf{u}_m\cdot\mathbf{t}$ add up to zero, one of them is ≥ 0, so one link must cover a plan distance of at least $\sqrt{(a^2/3 + |\mathbf{t}|^2)}$ and the sheets can be at most $\sqrt{(2a^2/3 - |\mathbf{t}|^2)}$ apart. So **the greatest spacing that taut links of length *a* allow is √(2/3) *a*, reached only with the sheet exactly in place**. A fabric whose sheets are pulled apart (by a frame, by spacers or by stuffing) therefore takes the exact geometry by itself, c or h alike; its accuracy is that of the link lengths and the node positions.

Cells and HexOct tiles. Our check: tiles.py of #75 at λ = −1/8, exact rationals Between two sheets, sheets and links close off three kinds of cell, one of each per node of a sheet: a tetrahedron on an up-triangle of the lower sheet (three nodes and the one node above linked to all three: 3 links), a tetrahedron under a down-triangle of the upper sheet (3 links), and an octahedron between a triangle below and the opposite triangle above (6 links). All edges have length *a*: the sheet threads and the links are the edges of the octet's regular tetrahedra and octahedra. At the weight λ = −1/8 of the HexOct walk (the weight of the ring build) the tile pieces of #75 are exactly these cells: the two tetrahedral pieces of a layer's tile are the tetrahedra standing on and hanging from that sheet (4 nodes each, all 6 edges *a*), and its two octahedral halves are the octahedra above and below cut at their mid-planes (3 nodes and 6 link midpoints each). Checked for layers of A-B-C, A-B, A-B-C-B (both layers) and A-B-C-A-C-B. The c/h difference of the tiles is the difference of the links: through an h sheet the octahedron above sits straight over the octahedron below; through a c sheet it is set aside. The mid-plane halving is part of the tile, not of the fabric: no yarn lies in the mid-planes. At other weights of the walk the tiles have other shapes (#75) and are not cells of the fabric.

2. How to knit it

(a) By hand: garter sheets with marked nodes, i-cord links

This is the spacer-fabric build done by hand: each sheet a firm knitted fabric whose nodes are marked stitches, each link a knitted cord from a node of one sheet to a node of the next. Double knitting is not suitable: its two layers have no gap, so the links could not have the length a (knitted-octet note, §2). The sheets close the tetrahedra and octahedra on their faces; the links are their slanted edges.

One sheet (3 × 3 nodes). Yarn: a firm, low-stretch yarn, for example mercerised cotton, DK weight, on 3.5 mm needles. Stitch: garter stitch (knit every row), which lies flat, does not curl and has little lengthwise give.

1. Swatch 30 stitches × 40 ridges; measure stitches per 10 cm (S) and ridges per 10 cm (R; one ridge = 2 rows).
2. Choose the node spacing a = ***N* = 16 stitches** and the node-row spacing ***M* ridges**, M the whole number nearest $0.866 \times 16 \times R/S$. For a square garter gauge ($R = S$), $M = 14$: the rows come out 14/16 = 0.875 of a instead of 0.866, +1.0%. (N must be even, so that every second node row can shift by half a spacing.)
3. Cast on 64 stitches ($4N$); work 42 ridges ($3M$, 84 rows); bind off.
4. Nodes: on the right-side row of ridges 7, 21 and 35 (counting from the cast-on), node row $j = 0, 1, 2$, the nodes are stitches $9 + 16i + 8j$ for $i = 0, 1, 2$ (counted from the start of the row). Knit each node stitch with the yarn and a 15 cm contrast thread held together, and leave the thread's ends hanging: it marks the node and guides the pick-up. The nine nodes form a 60° rhombus, node (i, j) at $i\cdot\mathbf{e}_1 + j\cdot\mathbf{e}_2$.
5. Knit all three sheets the same, and lay them the same side up and the same way round (cast-on edge towards you). The position A, B or C is not knitted into a sheet; it is set by the links (§1, self-centring).

One set of links (between two sheets).

1. Free length. The node centres are in the sheets' mid-planes; with fabric thickness t (garter in DK cotton about 3–4 mm, measure it), each end of a link runs $t/(2 \sin 54.74°) \approx 0.61\ t$ inside a sheet, so the cord between the two sheet faces is ℓ **= *a* − 1.22 *t*** long. Cut a card gauge of length ℓ and measure every cord on it under the same light pull.
2. On the upper face of the lower sheet, at a node, pick up and knit 3 stitches: the node stitch and its two neighbours in the ridge. Work a 3-stitch i-cord on two double-pointed 3.0 mm needles (knit 3, slide the stitches to the other end, pull the yarn across the back, repeat) until it matches the card. Count the rows of the first cord and work every other cord to the same count: equal counts make equal links. Put the 3 live stitches on a safety pin and tag the pin with the target node (table of §1: forward step (i, j), $(i-1, j)$, $(i, j-1)$; backward step (i, j), $(i+1, j)$, $(i, j+1)$). Leave out a link whose target is outside the 3 × 3 patch.
3. Work all links of this gap from the lower sheet. Lay the upper sheet over them, face up, and sew the 3 live stitches of each cord into its target node with a tapestry needle, working from above through the sheet (each stitch through the node stitch or a neighbour), so that each node of the upper sheet receives its links from below.
4. Then work the links of the next gap up from the upper face of this sheet. Each node of a middle sheet thus has up to 3 cords sewn in from below and up to 3 knitted out above. The links are knitted from the sheets' own stitches, so sheets and links are one continuous fabric; only the upper end of each cord is sewn.

Exact and approximate (hand). Exact: which node is linked to which, hence the c/h sequence and the structure through the thickness; the equality of the links (equal row counts). Approximate: the triangle of nodes (gauge; +1.0% in row spacing at $N = 16$, $M = 14$ with square gauge; garter also stretches crosswise, so the sheet must be held, see §4); the link length (i-cord stretches under load; a non-stretch core cord cut to length, with the i-cord knitted round it, fixes it); the node as a point (a node is a stitch, about 4.5 mm across at 22 stitches per 10 cm, against $a \approx 73$ mm). Open-mesh variant: knitting the sheet threads themselves as i-cords (6 meet at each node, 3 to knit per node) gives the octet's full strut network, edges of all cells as cords; exact in connections, about twice the cord work (3 sheet cords besides the 3 link cords per node and gap), and softer (knitted-octet note, §3).

(b) By machine: double-needle-bar spacer, one guide bar per link direction

From the knitted-octet note's sources [10, 11, 12] A warp-knitted spacer fabric on a double-needle-bar Raschel machine knits two ground fabrics at once, joined by pile yarns that travel across and are knitted into both; the pile yarns' lap and the bar distance set the spacing, and several guide bars allow several pile-yarn sets. Mapping (our design, untried): front ground = sheet k, back ground = sheet k+1, each knitted as an open net with its nodes on a triangular lattice; three pile guide bars, one per link direction. In needle and course units (wale spacing w, course spacing c), links 1 and 2 shog $\pm n_x$ needles with $a/2 = n_x w$ and advance n_y courses with $a/(2\sqrt{3}) = n_y c$; link 3 does not shog and goes back $2n_y$ courses. Both are possible at once only if $\boldsymbol{c/w = n_x/(\sqrt{3}\, n_y)}$, which is irrational for whole n_x, n_y and so is met only through the course density (take-down), approximately. A forward or backward step is the same machine set-up turned by 180° in plan, that is, the three pile bars lapping the opposite way. The bar distance is set so that each pile link has length a at slope 54.74°, which with the offset $a/\sqrt{3}$ gives the spacing $\sqrt{(2/3)}\, a$.

Sheet as a knitted triangular net (machine). Warp-knitted nets (the tulle and marquisette family) are standard; a net whose bars run in the three directions 0°, 60°, 120° with knots at the nodes would be made with pillar stitches along the wales for one direction and diagonal laps for the other two, so the same irrational ratio of course to wale spacing enters, and the net shrinks when it relaxes off the machine. Tolerances must be measured on a sample; we give none.

Exact and approximate (machine). Exact: the linking pattern (which node to which), repeated over the whole width; all links of one bar equal. Approximate: the triangle (course density, relaxation), the three bars' equal lengths (each bar has its own lap, so the three must be matched by trial). Limit: **one gap per machine run**. A Knitted HexOct needs every middle sheet to be the back of one gap and the front of the next. In the sources checked (knitted-octet note [7–16]) multilayer weft-knitted structures and 4-bed WholeGarment machines exist, but we found no machine knitting three or more *spaced* faces in one run. A multi-sheet machine fabric therefore needs either such a machine (not found) or hand joining of single-gap slabs, which doubles each shared sheet and is then no longer one continuous fabric.

3. Smallest test piece showing c against h

Three sheets, two gaps of links, 3 × 3 nodes per sheet, knitted twice. Only the middle sheet has a type (the outer sheets' types depend on sheets that are not there).

piece	word	steps	middle sheet	nodes	links (gap 1 + gap 2)	middle-sheet links per node (row j = 0, 1, 2)
T-c	A-B-C	forward, forward	**c**	27	42 (21 + 21)	4 5 4 · 5 6 5 · 4 5 4 (1 node with all 6)
T-h	A-B-A	forward, backward	**h**	27	42 (21 + 21)	6 6 4 · 6 6 4 · 4 4 2 (4 nodes with all 6)

Our check: scripts/check.py The first gap is the same in both pieces (links per bottom node 1 2 2 · 2 3 3 · 2 3 3); they differ only in the second gap. In T-c the top sheet's patch moves on by $(\mathbf{e}_1+\mathbf{e}_2)/3$ again; in T-h it returns over the bottom sheet. Seen along a row (Figure 1), T-c shows straight slants and T-h a chevron; from above, the octahedral cells of the two gaps of T-h sit one over the other, those of T-c do not.

Materials (per piece; yarn amounts are our estimate, about 1 cm of yarn per stitch in DK cotton). Three garter sheets of 64 stitches × 84 rows (about 5 400 stitches, about 55 m each); 42 i-cords of about 7 cm (about 1 m of yarn each with tails); together about 210 m, that is 3 balls of 50 g DK mercerised cotton (about 100 m each) per piece, 5 balls for both pieces. Needles: 3.5 mm pair, 3.0 mm double-pointed pair. Contrast thread for 27 node marks; 42 safety pins with tags; tapestry needle; card for the link gauge; optional non-stretch core cord (42 × ℓ plus knots); for holding: an embroidery hoop or a frame for each sheet, or a 60 mm spacer block, see §4. Sizes at 22 stitches per 10 cm: $a \approx 73$ mm, sheet spacing ≈ 59 mm, sheets about 29 × 19 cm (square garter gauge), piece about 13 cm thick.

Order of work. Knit 3 sheets → 21 cords up from sheet 1 → sew to sheet 2 → 21 cords up from sheet 2, targets by forward step (T-c) or backward step (T-h) → sew to sheet 3 → hold the sheets apart and measure the spacing (expect 59 mm) and the sideways position of the top sheet (T-c: moved on; T-h: over the bottom sheet).

4. Open points for DICTO

- **Holding the shape.** The fabric is exact only pulled apart (§1). By frame (hoops or a rigid border on each sheet), by stuffing each cell, by spacer blocks, or by stiffening the cords (starch, glue, a wire or nylon core).
- **Scale.** $N = 16$ stitches gives $a \approx 73$ mm in DK cotton; larger a means more exact nodes, more yarn. $N = 30$, $M = 26$ gives the row spacing to 0.07% but $a \approx 14$ cm.
- **Sheet: solid or open.** Solid garter sheets (this spec), or open triangular nets so that the cells can be seen through.
- **Machine route.** Whether to look further for a machine that knits three or more spaced faces, or to keep the machine to one-gap samples.

Conclusion. The Knitted HexOct is exact in its connections and self-centring in its geometry: every link has length a, straight through c sheets and folded at h sheets, and at $\lambda = -1/8$ its cells are the pieces of the HexOct tiles. **Status:** specification and exact link checks done; no piece knitted yet; next step the two hand test pieces T-c and T-h.

Sources

[1] Knitted Octet, research note 2026-11-02 (studies/2026-11-02-knitted-octet/paper.html), §§1–3 and its sources [7]–[16] (spacer fabrics: US 6779369, US 5385036, EP 0748889, Karl Mayer double-needle-bar Raschel notes; multilayer knits: US 6244077, TU Dresden; Shima Seiki WholeGarment). Used for the spacer-fabric facts in §2(b); no new sources were searched for this spec.
[2] DICTO, HexOct: Two Tiles for Every Stacking (#75): tiles, weight walk; tiles.py (studies/2026-10-29-endless-stack-cell).
[3] DICTO, Rule of Halves (#57). [4] DICTO, HexOct: Knitting Crystals (#83), DICTO DOI: 10.5281/zenodo.23848271: the edge-on slice as a knitted chart (slant and chevron).
[5] HexOct ring build, study 2026-10-31-hexoct-ring-build (the weight $\lambda = -1/8$).

Knitted HexOct: is the c/h stacking a design dial?

Lattice mechanics and a see-through/airflow proxy for c, h and mixed stacking sequences



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In the Knitted HexOct, which DICTO proposed, each knitted sheet is one close-packed layer of nodes. Links join every node to the 3 nearest nodes of the next sheet. At a c sheet the links run straight on; at an h sheet they fold back in a chevron. All link lengths, angles and node counts are the same whatever the stacking, so all stackings have the same density. That makes the stacking sequence a candidate *design dial*: the same yarn and the same stitch count, but a different property, depending only on where the links turn. This study checks that idea in idealised lattice models.

> **Result.** Through-thickness compression, buckling onset and rigidity are the same in every sequence (to 4 digits). On those properties c/h is no dial. Two properties do change strongly: (1) the **straight see-through area** along the stacking axis is 34.7 % in 2H, 9.9 % in 4H and 3.25 % in 3C at link radius 0.05 *a*, with equal porosity (10.7×). At radius 0.1 *a* it is 14.3 % in 2H and 0 in 3C. (2) **Free transverse shear stiffness** changes when the sheets are soft in-plane: 3C/2H = 0.057 (17.5×) with links only and rigid joints. Graded stacks fall in between. The ratio rises to 0.900 when the sheets are as stiff as the links. The controlling number for shear is the stack's net c-twist (mean Hägg symbol), not its share of c sheets.

1 Model

Geometry (from the spec, studies/2026-11-02-knitted-hexoct): node spacing $a = 1$, sheet spacing $\sqrt{(2/3)}$, links of length 1 at 54.7356°. Each block has 12 sheets with 6 × 6 nodes per sheet and is periodic in-plane. A program checks every node: 3 links up, 3 links down and 6 in-sheet neighbours. The sequences are written as c/h labels of the 10 inner sheets: 3C (all c), 2H (all h), 4H (hc…), 6H (hcc…), Fibonacci (abaababaab, a → c, b → h), and two graded stacks (5 h at the bottom and 5 c at the top, and the reverse). **Members** are round rods, radius $r = 0.05\ a$, $E_s = 1$, $\nu = 0.3$. They are modelled as pin-jointed bars and as rigid-jointed 3D Euler–Bernoulli frame members. There are three sheet variants: *links only* (the sheets are just nodes); *stiff sheets* (in-sheet triangular members 100× stiffer than the links, our stand-in for nearly rigid sheets); and *full* (in-sheet members the same as the links; 3C is then the octet truss, 2H the hcp truss). **Tests on the block:** compression with bonded platens (C33) and with lubricated platens and free lateral strain (E3); transverse shear with bonded platens and with free in-plane strain (the bulk limit is $1/S_{55}$); and in-plane E1, E2 and G12 of the free plate. The 3D-periodic 6×6 stiffness matrix C is computed for 3C, 2H, 4H and 6H. **Check:** for 3C (full, pin joints) the computed E[100] equals $\bar{\rho}/9$, the octet-truss value of Deshpande, Fleck & Ashby (2001). The computed ratio is 1.0000000000. **Energy proxy:** an event-to-event analysis with pin joints. Each strut buckles at the pinned Euler load $\pi^2 E_s I/a^2$ and then carries that load unchanged; tension stays elastic. We report the work done up to 5 % strain. **Airflow proxy:** rods are capsules (round ends). The straight see-through fraction is exact: a line along the stacking axis misses a capsule exactly when its plan point lies more than r from the capsule's plan segment. Tortuosity is the shortest open path from each open point of the top plane to the bottom plane, divided by the thickness. It is computed on a 26-neighbour voxel graph, h = 0.02 *a*, checked at 0.0125 *a*.

2 Properties per sequence

sequence	word	(Hägg)	C33 bonded	E3 free	G13 bonded	G13 free links only	G13 free full	E1 in-plane links only	work 5 % shear xz	see-through r = 0.05	see-through r = 0.1	tortuosity r = 0.05 / 0.1
3C	ABCABCABCABC	1.000	14.865	0.492	3.715	0.193	3.360	0.051	3.50	3.25 %	0	1.003 / 1.039
2H	ABABABABABAB	0.091	14.865	0.492	3.715	3.378	3.699	0.063	3.28	34.7 %	14.3 %	1.001 / 1.008
4H	ABACABACABAC	−0.091	14.865	0.492	3.715	3.283	3.699	0.055	3.28	9.9 %	0	1.002 / 1.021
6H	ABACBCABACBC	−0.091	14.865	0.492	3.715	3.276	3.699	0.055	3.28	3.25 %	0	1.003 / 1.033
Fibonacci	ABCBACABACBC	−0.091	14.865	0.492	3.715	3.275	3.699	0.055	3.28	3.25 %	0	1.003 / 1.032
graded c top / h bottom	ABABABACBACB	−0.455	14.865	0.492	3.715	0.927	3.626	0.057	3.28	3.25 %	0	1.003 / 1.022
graded h top / c bottom	ABCABCACACAC	0.455	14.865	0.492	3.715	0.927	3.626	0.057	3.28	3.25 %	0	1.003 / 1.027

Stiffness in 10^{-3} E_s. C33, E3 and G13 bonded: rigid joints, links only (relative density $\bar{\rho}$ = 0.0305). These values are the same for all sequences in every variant; with in-sheet members they are 14.810–14.865, 13.44–13.52 and 3.70–3.76 ($\bar{\rho}$ = 0.0639). "Full": pin joints, links + in-sheet members. Work done in xz shear up to 5 % strain, in 10^{-6} E_s, full, pin joints. In compression the first buckling happens at strain 0.0093 (bonded) and 0.0102 (lubricated), at stress 1.370×10^{-4} E_s, in every sequence; there all 1188 links buckle together. Porosity 0.9406 (r = 0.05) and 0.7869 (r = 0.1) in every sequence. Full table: data/tables.txt.

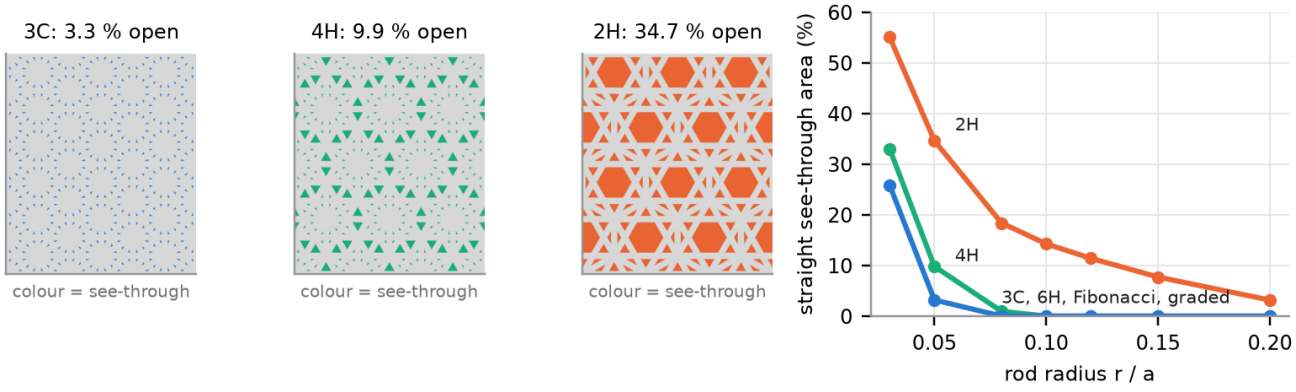


Figure 1. Straight see-through area along the stacking axis. Left: open plan area (colour) over 3 × 2 rectangular cells at link radius 0.05 *a*. Right: open area against rod radius. The open area depends only on which of the three link plan pairs AB, BC, CA occur: 2H has one, 4H two, and 3C, 6H, Fibonacci and the graded stacks have all three.

3 Rigidity: what h does

Links only, pin joints. Every node has 6 bars, so the periodic cell has exactly 3N bars, the Maxwell isostatic count, in every sequence. The compatibility matrices show the same picture in every sequence: 3 macroscopic strains cost no bar stretching. These are the in-plane strains combined with $\varepsilon_{zz} = -(\varepsilon_{xx}+\varepsilon_{yy})/2$; in 3C they are mixed with transverse shear. The 6×6×12 periodic cell also has periodic mechanisms: 33 (3C), 37 (4H), 39 (2H), 51 (6H). Each comes with exactly as many self-stress states plus 3, as Guest & Hutchinson (2003) show must happen in an isostatic repetitive frame. The finite 12-sheet block has 105 mechanisms in every sequence. So the link net alone is not rigid, whether c or h. It carries through-thickness compression only between bonded faces. **With in-sheet members** every sequence is rigid (0 mechanisms, 1191 self-stress states in the block). The hcp-type h stack is as rigid and stretch-dominated as the octet-type c stack: C33, C13 and C44 are equal, and E3 is identical.

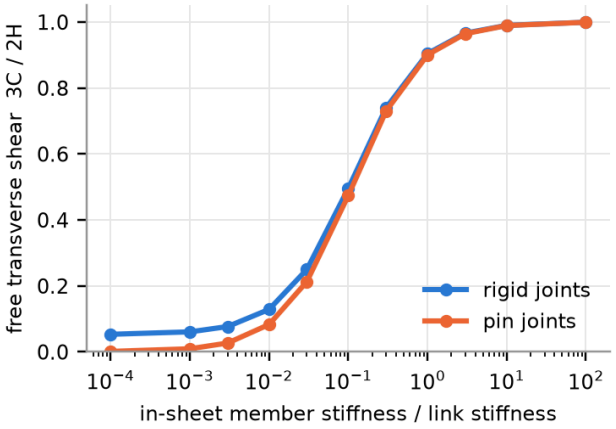


Figure 2. Shear dial strength (periodic bulk, $1/S_{55}$ of 3C over 2H) against the stiffness of the in-sheet members relative to the links. Soft sheets give a strong dial (0.05 rigid joints, 0.001 pin joints at 10^{-4}). Sheets as stiff as the links give 0.90, and 100× stiffer sheets 0.999.

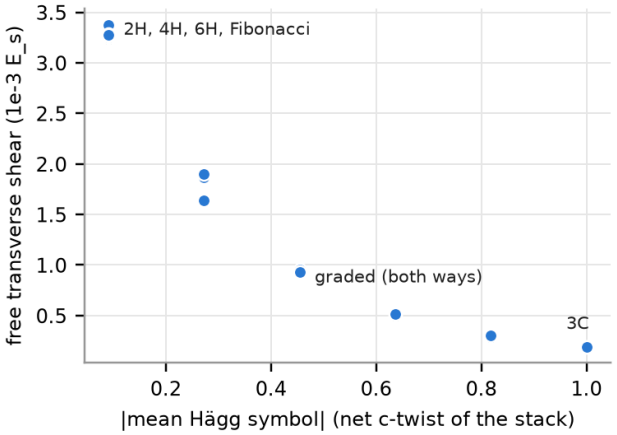


Figure 3. Free transverse shear stiffness (links only, rigid joints, 12-sheet block) against |mean Hägg symbol| for the 7 sequences, 11 bottom-c/top-h stacks and 5 stacks of c runs split by single h sheets. An h sheet reverses the twist sense, so c runs split by an h cancel: ccccchcccc gives 3.24, close to 2H.

4 Why shear changes and compression does not

The c stack has 3-fold (trigonal) symmetry about the stacking axis. Its stiffness matrix therefore has a shear-stretch term $C_{14} = 1.31\times10^{-3}\ E_s$: transverse shear is coupled to in-plane stretching and in-plane shear. In 2H, 4H and 6H, $C_{14} = 0$, as hexagonal symmetry requires. When the faces are free to strain in-plane, the 3C fabric can give way in shear by straining its sheets. If the sheets are soft, that costs little, so $1/S_{55}$ drops well below C_{55} (roughly by coupling² / in-plane sheet stiffness). With bonded platens the in-plane strain is blocked and every sequence gives the same value. Through-thickness compression loads all links equally (each at $\cos^2 = 2/3$ of the strain) whether they run straight on or fold back. So C33, E3 and the buckling onset do not depend on the stacking.

5 Conclusion and possible uses

Yes, c/h is a usable dial, for two properties only. It sets the straight see-through area: 10.7× from 3C to 2H at $r = 0.05\ a$, at equal porosity, mass and thickness. It also sets the free transverse shear stiffness, by up to 17.5× when the sheets are soft and laterally free. It does **not** set compression stiffness, buckling onset, compression energy or pin-jointed rigidity. Possible uses, with what the model does and does not support:

- **Insulation and breathable layers:** a c-rich stack closes the straight sight-lines through the thickness, and a 2H stack keeps one third of the area open. This is a line-of-sight result, relevant to radiation and light passing straight through. We did not compute heat transfer or permeability. Tortuosity barely changes (1.001–1.039).
- **Padding and footwear:** cushioning stiffness in compression is not changed by c/h in this model. Shear compliance (sole and heel shear, comfort layers) is changed, if the knitted sheets are much softer in-plane than the links, as knits usually are. 3C couples shear to in-plane strain; graded stacks give intermediate shear stiffness through the thickness.
- **Composite preforms:** once resin stiffens the sheets, the shear dial falls to about 10 % (ratio 0.900 at equal stiffness). The open straight z-channels of 2H could matter for through-thickness infiltration, but no flow solve was made.

Model limits. Idealised geometry: straight rods meeting at points, every link exactly length a. No yarn mechanics: no loops, compaction, friction, yarn twist, or yarns that cannot carry compression. Linear elasticity, apart from the buckling plateau of the energy proxy; no post-buckling softening; no contact. A finite block of 12 sheets, periodic in-plane. Results are given relative to the rod modulus E_s and depend on r/a. Airflow is a geometric proxy (straight see-through area and shortest-path tortuosity), not a permeability. Physical tests of knitted samples (T-c and T-h of the spec) are needed before any application claim.

6 Prior art and sources checked (2 November 2026)

Octet-truss stiffness and the stretch-dominated/bending-dominated distinction are from Deshpande, Fleck & Ashby [1, 2]; our 3C check reproduces [1]. The mechanism and self-stress counts follow Guest & Hutchinson [3]. Spacer fabrics with one gap are established cushioning materials [4]. 3D woven, braided, stitched and knitted preforms are reviewed in [5, 6]. Lattices that copy crystal grains and twins ("meta-grains") were made by Pham et al. [7]. In atomic polytypes the elastic constants vary almost linearly with hexagonality [8]. In the results we saw, we found no mechanical comparison of hcp-type with fcc-type (octet) truss lattices, and none of mixed c/h stackings as a design variable. This is not a full literature review.

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3. S. D. Guest, J. W. Hutchinson, On the determinacy of repetitive structures, J. Mech. Phys. Solids 51 (2003) 383–391. Record seen.
4. Y. Liu, H. Hu, L. Zhao, H. Long, Compression behavior of warp-knitted spacer fabrics for cushioning applications, Textile Res. J. 82 (2012) 11–20; also Y. Liu, H. Hu, TRJ 86 (2016) 3–12. Abstracts and records seen.
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7. M.-S. Pham, C. Liu, I. Todd, J. Lertthanasarn, Damage-tolerant architected materials inspired by crystal microstructure, Nature 565 (2019) 305–311. Records seen.
8. M. Keller et al., First-principles insight in structure-property relationships of hexagonal Si and Ge polytypes, arXiv 2302.10822 (2023). Library copy, pp. 1 and 11 read.

Also checked, not cited: library: Lalvani 1990, NASA TM-102635 (layered oct-tet truss part counts; pp. 1–3 read; no mechanics); Goyal & Lakes 2026 and Korshunova et al. 2020/2021 (octet truss; catalogue entries only); Krishnamurthy, Akleman et al. 2020/2022 (woven tiles; snippets only); Rodriguez-Horta et al. 2017 (close-packed layer interactions; catalogue entries only). Library queries: octet truss stretch-dominated; spacer fabric compression; 3D woven/knitted preform; hcp/fcc stacking elastic constants polytypes; Maxwell isostatic periodic lattice mechanisms. Web queries: the Deshpande–Fleck–Ashby octet paper; Guest–Hutchinson determinacy; warp-knitted spacer fabric compression; the Mouritz 1999 review; "hcp truss lattice vs fcc octet truss lattice stiffness stacking sequence"; the Pham 2019 meta-crystals paper. Scripts, data and full tables: study folder 2026-11-02-hexoct-knit-mechanics.